

This assignment will not be graded and is only for practice.

Level 1

1) Which of these sets form an orthonormal basis of $\mathbb{R}^3$ with respect to the dot product as the inner product in $\mathbb{R}^3$? **1 point**

- ☐ $\{(1, 0, 0), (0, 1, 0)\}$
- ☐ $\{(2, 0, 0), (0, 2, 0), (0, 0, 2)\}$
- ☐ $\{(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}), (\frac{-3}{\sqrt{11}}, \frac{1}{\sqrt{11}}, \frac{1}{\sqrt{11}}), (\frac{1}{\sqrt{66}}, \frac{7}{\sqrt{66}}, \frac{-4}{\sqrt{66}})\}$
- ☐ $\{(2, 1, -1), (-1, -1, 1), (1, 2, -2)\}$
- ☐ $\{(\frac{1}{\sqrt{2}}, 0, \frac{-1}{\sqrt{2}}), (0, 1, 0), (\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\{(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}), (\frac{-3}{\sqrt{11}}, \frac{1}{\sqrt{11}}, \frac{1}{\sqrt{11}}), (\frac{1}{\sqrt{66}}, \frac{7}{\sqrt{66}}, \frac{-4}{\sqrt{66}})\}$$

$$\{(\frac{1}{\sqrt{2}}, 0, \frac{-1}{\sqrt{2}}), (0, 1, 0), (\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})\}$$

2) Consider two orthogonal vectors a, b in $\mathbb{R}^2$. If $a + b$ and $a - b$ are orthogonal, then choose the correct option. **1 point**

- ☐ $\|a\| = \|b\| = 1$
- ☐ $\|a\| = \|b\|$
- ☐ $\|a\| = 2 \|b\|$
- ☐ $2 \|a\| = \|b\|$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\|a\| = \|b\|$$

3) Consider $a = (1, 1)$, $b = (1, -1)$. Let $V = \text{Span}\{a, b\}$. Choose the correct options by considering the standard inner product (dot product).

1 point

- ☐ Vectors a, b form an orthogonal basis for V .
- ☐ Vectors a, b form an orthonormal basis for V .
- ☐ There exist scalar multiples of a, b which form an orthonormal basis
- ☐ There do not exist scalar multiples of a, b which form an orthonormal basis

No, the answer is incorrect.

Score: 0

Accepted Answers:

Vectors a, b form an orthogonal basis for V .

There exist scalar multiples of a, b which form an orthonormal basis

Level 2

4) Choose the set of correct statements.

1 point

- ☐ The determinant of a matrix formed by 3 orthonormal vectors in $\mathbb{R}^3$ is ± 1 .
- ☐ The determinant of a matrix formed by 3 orthonormal vectors in $\mathbb{R}^3$ is 0.
- ☐ The determinant of a matrix formed by 2 orthonormal vectors in $\mathbb{R}^2$ is ± 1 .
- ☐ The determinant of a matrix formed by 2 orthonormal vectors in $\mathbb{R}^2$ is 0.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The determinant of a matrix formed by 3 orthonormal vectors in $\mathbb{R}^3$ is ± 1 .

The determinant of a matrix formed by 2 orthonormal vectors in $\mathbb{R}^2$ is ± 1 .

5) Consider a system of linear equations:

1 point

$$2x_1 + 2x_2 + 7x_3 = b_1$$

$$2x_1 + x_2 - 10x_3 = b_2$$

$$3x_1 - 2x_2 + 2x_3 = b_3.$$

Let A be the coefficient matrix of the given system of linear equations. Let a matrix B contain the column vectors of A , which are normalized by their respective norms, as its columns (i.e. first column vector of A normalized by its norm is the first column of B). Which of the following statements are true ?

- ☐ The determinant of BB^T is 1.
- ☐ BB^T is an identity matrix.
- ☐ BB^T is a scalar matrix.
- ☐ BB^T is a diagonal matrix.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The determinant of BB^T is 1.

BB^T is an identity matrix.

BB^T is a scalar matrix.

BB^T is a diagonal matrix.

6) Choose the correct option(s).

1 point

- ☐ The vectors in an orthonormal set are linearly independent.
- ☐ A set of linearly dependent vectors can be orthonormal.
- ☐ A set of linearly independent vectors is always orthonormal.
- ☐ A set of linearly independent vectors is always orthogonal but not orthonormal.
- ☐ The vectors in an orthogonal set are linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The vectors in an orthonormal set are linearly independent.

The vectors in an orthogonal set are linearly independent.

Consider the inner product $\langle u, v \rangle = 3u_1v_1 + 2u_2v_2$ where $u = (u_1, u_2)$, $v = (v_1, v_2)$ are vectors in $\mathbb{R}^2$. Answer questions 7 and 8 based on this information.

7) Which of the following is an orthonormal basis with respect to the inner product defined above? **1 point**

- ☐ $\{(2, 3), (-4, 4)\}$
- ☐ $\{\frac{1}{\sqrt{30}}(2, 3), \frac{1}{\sqrt{80}}(-4, 4)\}$
- ☐ $\{\frac{1}{\sqrt{13}}(2, 3), \frac{1}{\sqrt{32}}(-4, 4)\}$
- ☐ $\{(2, 3), (-3, 2)\}$
- ☐ $\{\frac{1}{\sqrt{13}}(2, 3), \frac{1}{\sqrt{13}}(-3, 2)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\{\frac{1}{\sqrt{30}}(2, 3), \frac{1}{\sqrt{80}}(-4, 4)\}$$

8) Use the orthonormal basis $\{u, v\}$ obtained in question 7 with respect to the defined inner product. Express the vector $(4, 0)$ as a linear combination of the basis vectors u and v , as $(4, 0) = c_1u + c_2v$. Which of the following gives the coefficients of the linear combination? **1 point**

- ☐ $c_1 = \frac{24}{\sqrt{30}}, c_2 = \frac{-48}{\sqrt{80}}$
- ☐ $c_1 = \frac{24}{\sqrt{13}}, c_2 = \frac{-48}{\sqrt{32}}$
- ☐ $c_1 = \frac{8}{\sqrt{13}}, c_2 = \frac{-16}{\sqrt{32}}$

☐ $c_1 = \frac{24}{\sqrt{30}}, c_2 = \frac{48}{\sqrt{80}}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$c_1 = \frac{24}{\sqrt{30}}, c_2 = \frac{-48}{\sqrt{80}}$

Your score is: 0/8